

POLS 158/PUBPOL 158/358G

AI & Politics

Simone Paci, PhD
Yiqing Xu, PhD

Who Are We?

Simone Paci *see-moh-neh pha-chee*

Lecturer & Associate Director

CDDRL

Education: Yale, Columbia (2023)

Research:

- Causal inference
- Political Economy
- Comparative Politics

Yiqing Xu */ee-ching shü/*

Associate Professor of Polisci

Stanford Causal Science Center, SCCEI

Education: Fudan, PKU, MIT (2016)

Research:

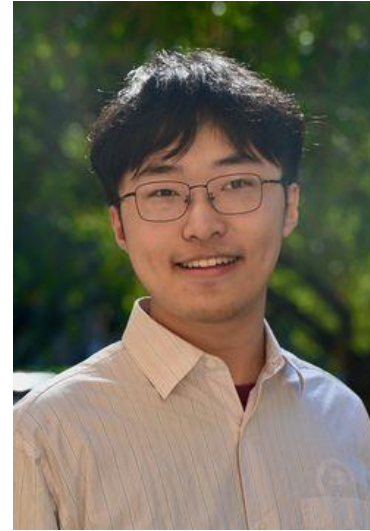
- Causal inference
- AI for social science research
- Comparative Politics

Two Words From Our Wonderful TAs

Xinru Pan



Zihan Xie



Introduce Yourself to Your Neighbor

- 1) Name
- 2) Where are you from?
- 3) Where & what do you (plan to?) study?
- 4) When did you first use AI? What did you think?

We will ask some of you to introduce your neighbor to the class!

Roadmap

1 Goals, Expectations, Structure

2 The Politics of AI

3 AI for Politics

Part I: Class Info

The Politics of AI: Goals

1. Explain core concepts behind modern AI technology
2. Use AI tools effectively and responsibly
3. Analyze how AI impacts power, resources, legitimacy
4. Assess emerging empirical trends in the politics of AI
5. Design and execute a project using or studying AI & Politics
6. Present and critique technical and/or analytical work on AI

The Politics of AI: Expectations

Participation & Prep

- class/section activities
- Weekly prep

Graded work

- 8 homework projects
- 3 project milestones
- 3 peer reviews

Presentation:

- Final Presentation (end-of-quarter)

Category	Weight
Participation & HW	35%
Presentation	15%
Project & Milestones	35%
Peer Review	15%

Projects

Data-Driven Paper

- a. Problem → Research Question
- b. Argument → Hypothesis
- c. Design → Data Collection → Analysis

Three Milestones

- 1. Proposal & Design
- 2. Implementation & Preliminary Results
- 3. Final Paper

Any “AI & Politics” Topics! Be curious & ambitious!

Section Signup on Canvas

Stanford University





Account



Dashboard



Courses



Calendar



Inbox



History



Help

Fall 2026

Home

Assignments

Discussions

Grades

People

Syllabus

Collaborations

SCRIBE

Zoom

Panopto Course Videos

Everyone

Groups

+ Group

Q Search Groups or People

Section 1: Xinru Thursday 10:30 am Discussion Sections	0 students	Not joinable in Student View
▶ Section 2: Xinru Thursday 11:30 am Discussion Sections	1 student	Not joinable in Student View
▶ Section 3: Zihan Friday 9:30 am Discussion Sections	5 students	Not joinable in Student View
Section 4: Zihan Friday 11:30 am Discussion Sections	0 students	Not joinable in Student View

Technology Use & AI Policy

This class incorporates AI in a carefully studied way:

- 1) Insulation
- 2) Augmentation
- 3) AI-skills development

AI & Education is an ongoing conversation:

- Offloading the creative struggle
- Cognitive sovereignty

Grading

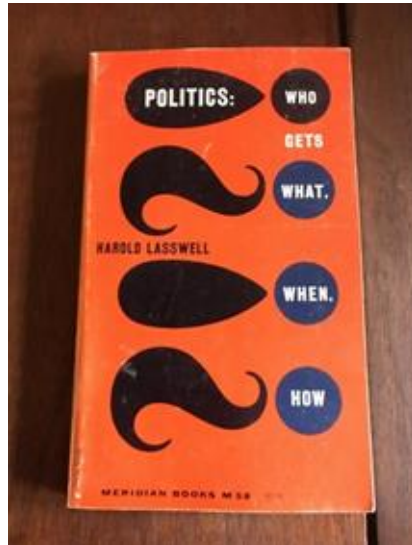
- Final Project (65%) – single or two-person group
 - Two milestones (20%)
 - Final paper (30%)
 - Presentation (15%)
- Section Activities (25%)
 - Pass/No-Pass
- Peer Review (10%)

Questions?

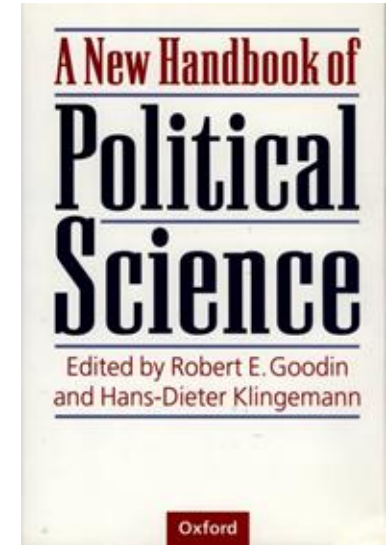
Part II: The Politics of Revolutionary Technological Change

What is Politics?

Lasswell (1950)
“Who gets what,
when and how”



Goodwin (2009)
“the constrained use
of social power”



Constrained Social Power → Distribution

Bringing Actors In

Actors = “anyone/anything that has agency”

→ make decisions over a menu of possible actions

→ have a utility function (preferences, incentives)

→ have beliefs

Who are actors in the “Politics of AI”?

Who can make consequential decisions?

What are their incentives?

What are their beliefs?

Bringing Institutions In

Institutions

- formal (e.g. state institutions, market structure...)
- informal (e.g. culture, norms...)
- “rules of the game” = constraints & opportunities

What are the important institutions in the “Politics of AI”?

Are they formal or informal?

How do they shape the actors’ menu of actions?

Can Technology be Political?

Perhaps technology is epiphenomenal...?

= simply reproduces the status quo politics



At least 2 ways of changing politics (Winner, 1980):

1. External shock to the status quo
2. Inherently political technologies

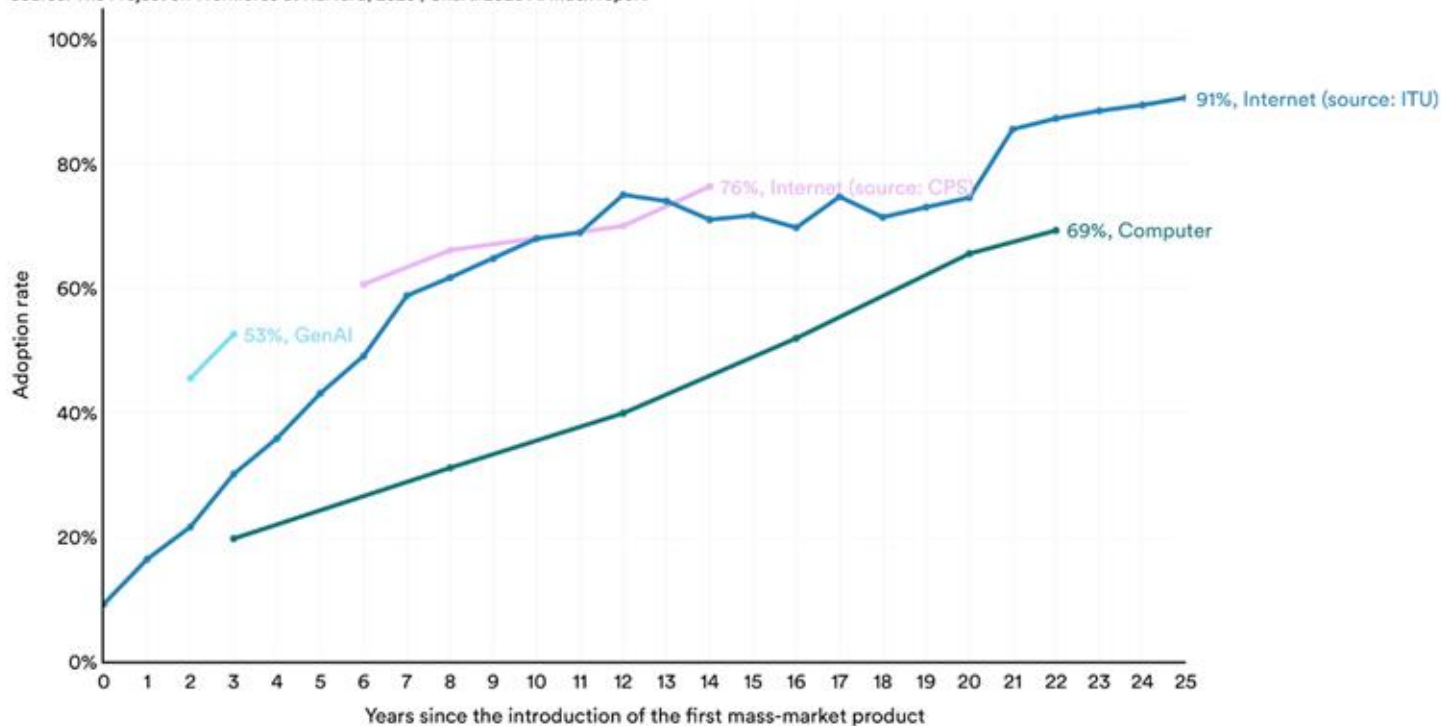
New Technological Order = Winners & Losers

1. External shocks → changes in distribution of resources
2. New incentives → actors react to the change
3. Institutions → provide menu of options

The speed of this new tech is unprecedented

Speed of AI adoption by technology

Source: The Project on Workforce at Harvard, 2025 | Chart: 2026 AI Index report



G.P.T.'s

Generative Pre-trained Transformer

But before then...

General Purpose Technology

Inherently Political Technologies

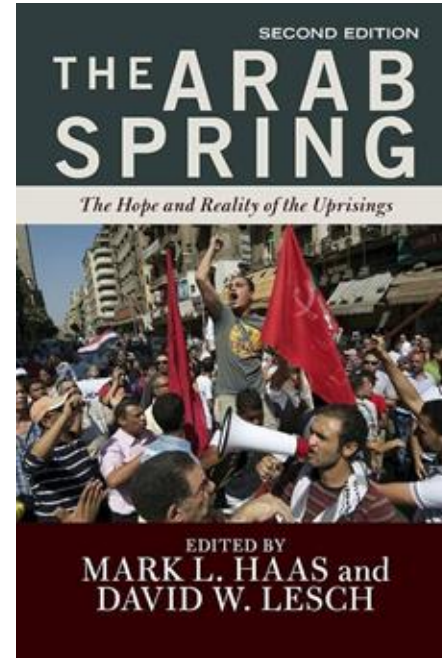
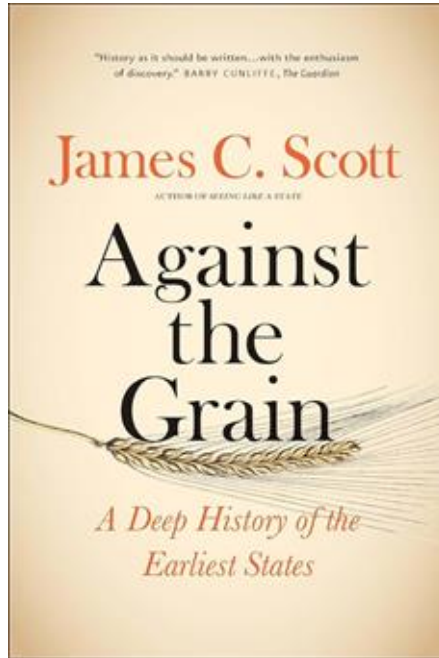
If a technology impacts society as whole, one question is:

Decentralizing / democratizing power

vs.

Centralizing / concentrating power

Inherently Political Technologies?



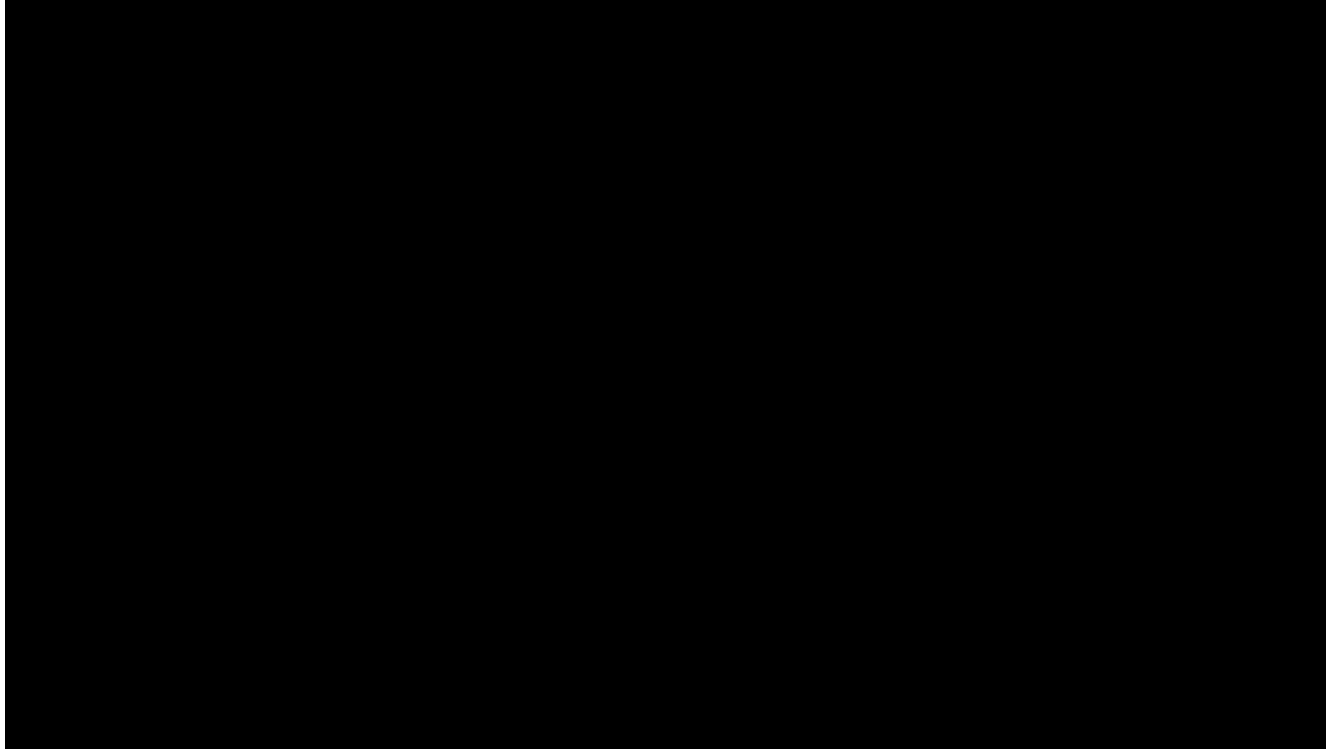
The Impact of Inherently Political Technologies

Need to understand both:

1. Political implications of adoption / diffusion
2. Nature & functioning of the technology itself

Part III: Artificial Intelligence: The Tech

Motivation



<https://www.youtube.com/watch?v=zuZ2aotrJs>

Ilya Sutskever

Computer scientist
inventor of AlexNet and
CNN, former OpenAI
cofounder

Commencement Speech at
University of Toronto
Jan 6, 2025

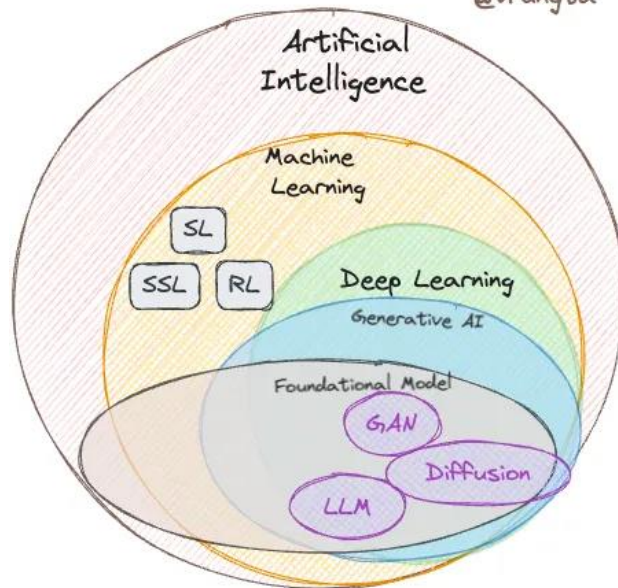
The Big Picture

- Artificial Intelligence (AI): Algorithms and systems that mimic human reasoning and decision-making.
- Machine Learning (ML): AI methods that learn predictive rules or recognize patterns directly from data.
- Deep Learning (DL): ML using multi-layer neural networks to model complex patterns.
- Generative AI (GenAI): Models that learn data distributions to create new content.
- Large Language Models (LLMs): Transformer-based GenAI trained on massive text to generate and understand language.

AI -> ML -> DL -> GenAI, FM -> LLM

*These are broad categorization, industry does not have alignment on some overlaps

@vrungta



LLM - Large Language Model

RL - Reinforcement Learning

SL - Supervised Learning

SSL - Self Supervised Learning

G,AN - Generative adversarial network

Diffusion - Stable diffusion etc models

Source: <https://vrungta.substack.com/>

AI & DL by Tasks (examples)

- **Natural Language Processing (NLP)**: Enables computers to understand, interpret, and generate human language.
- **Computer Vision (CV)**: how computers can be made to gain high-level understanding from digital images or videos.
- **Robotics**: the design, construction, operation, and use of robots to perform tasks.
- **Speech Recognition**: identify and convert spoken language into written text or vice versa.
- **Science**: the application of computational intelligence to analyze data and make new discoveries across various scientific fields.

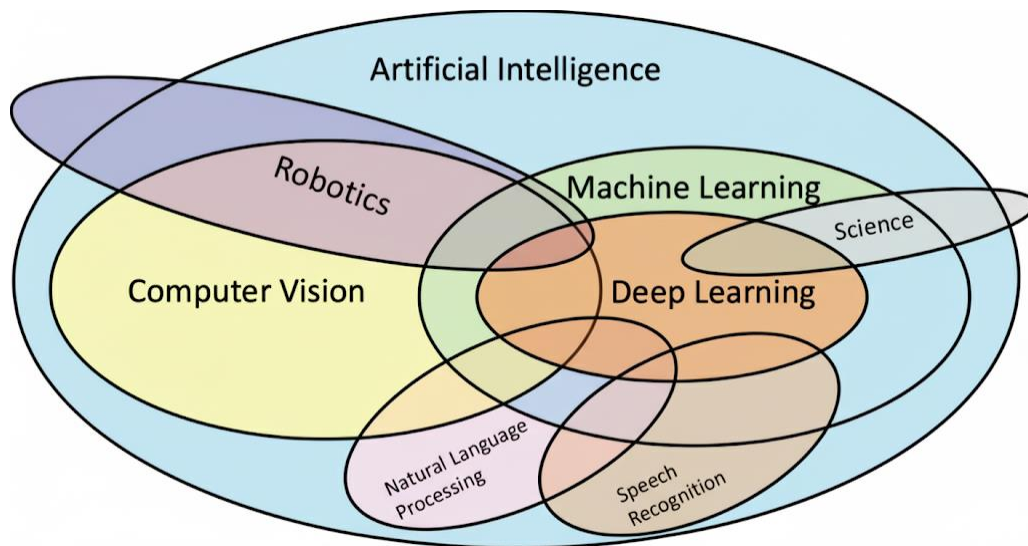


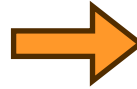
Figure credit: Fei-Fei Li & Yuan Yuan

Learning Goals for Track II

Have a sophisticated but non-technical understanding of popular AI architecture

Communicate effectively with people who have a technical background

Feel comfortable using existing AI tools



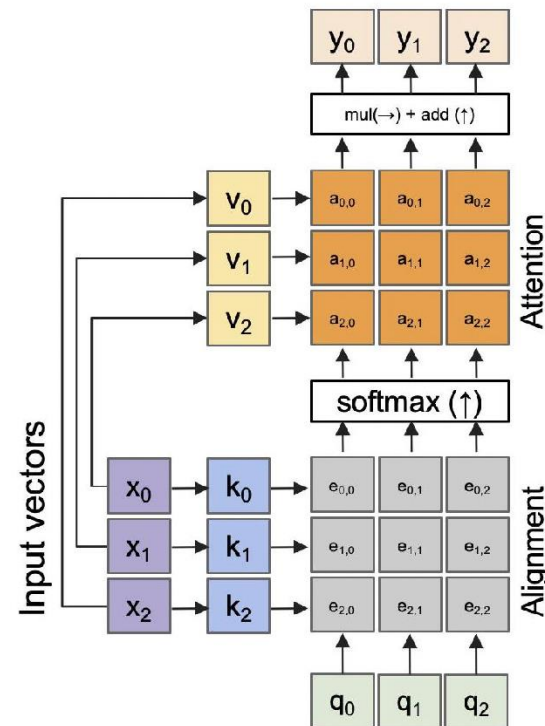
Make AI a powerful, well-rounded daily assistant

Understand the strength and limitations of current AI technology

Adapt AI tools to your social (or business) applications

Demystifying the AI World

- Demystifying AI vocabulary and ML concepts
 - SGD, RLHF, RNN, GAN, Transformers
- Applying existing AI tools
 - ChatGPT/Gemini/Claude
 - Identifying challenges of these tools
- Developing your personal AI routine
- This class is not about
 - Developing technical skills of building AI algorithms
 - A deep dive into AI applications of a particular scientific domain



Social and Political Applications of AI

- Find your **domain** — What area interests you the most?
- Locate the **problem** — What bother people in this domain? Is it scalable? How large is the “market”?
- Define the **tasks** — What should AI solve?
- Analyze technical challenges — What **AI architecture** should be used (ask Claude/Codex)?
- Execution and build a fast feedback loop



Reminders:

- Section Signup
- Start thinking about your projects

See you next class